1.Executive Summary

This report presents a comprehensive predictive analytics project utilizing the 1994 US Census Income dataset, which consists of 32,561 records and 15 variables. The primary objective of this study is to develop a robust binary classification model capable of predicting whether an individual earns more than $50,000 annually based on 14 demographic, educational, and occupational attributes.

To achieve this, the project adheres to the six phases of the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology. The entire analytical workflow was developed and executed within the KNIME Analytics Platform. The data preparation phase included critical exploratory data analysis using the Statistics node, missing value imputation, data filtering, and class balancing to ensure unbiased model training.

During the modeling phase, four distinct machine learning algorithms were trained and evaluated: Decision Trees, Random Forest, Artificial Neural Networks-MLP, and Logistic Regression. The performance of each model compared using multiple evaluation metrics, including Accuracy, Sensitivity, Specificity, and Area Under the ROC Curve (AUC).

Furthermore, the study delves into the interpretability of the models by analyzing the first three levels of the Decision Tree and extracting a Variable Importance graph from the Random Forest statistics. This report provides a detailed comparative analysis of the implemented models, identifying the optimal algorithm for this specific predictive task and offering insights into the most significant socioeconomic factors affects income levels.

2. Business Understanding

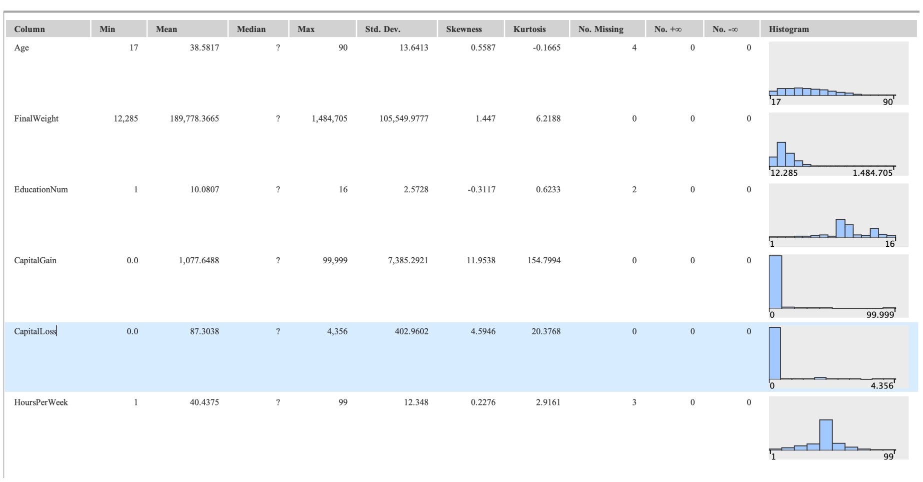
The primary objective of this project is to solve a classification problem using the 1994 US Census Income dataset. The core business goal is to build machine learning models that can accurately predict whether an individual earns more than $50,000 a year based on their personal and professional details.

Predicting income levels is highly valuable for real-world business applications. For example, financial institutions can use this information for credit risk assessment, while marketing agencies can use it to target the right demographic groups for premium products.

To achieve this goal, the project uses the KNIME Analytics Platform to explore the data, prepare it for analysis, and train four different predictive algorithms (Decision Trees, Random Forest, Artificial Neural Networks, and Logistic Regression). The ultimate success criterion for this project is to identify the most accurate and reliable model for this specific income classification task.

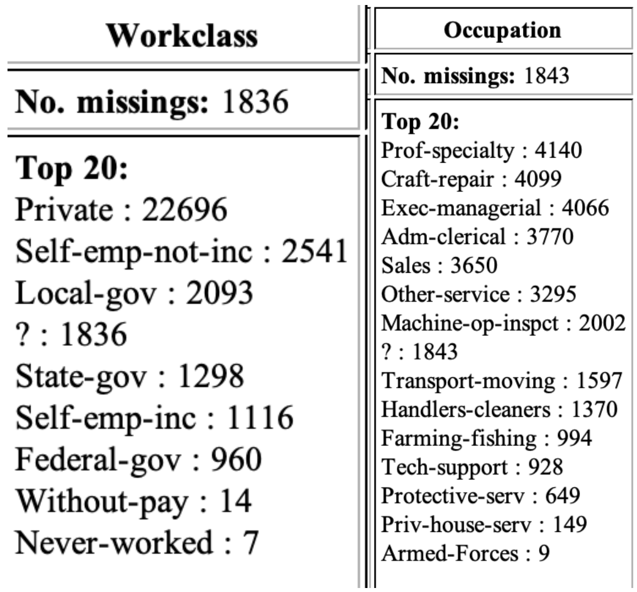
3. Data Understanding

In this phase, the Statistics node in KNIME was utilized to comprehensively examine the properties, distributions, and data quality of the Adult Income dataset. Understanding the underlying structure of the data is a critical prerequisite before moving on to the data preparation and modeling phases.



Column	Min	Mean	Median	Max	Std. Dev.	Skewness	Kurtosis	No. Missing	No. +∞	No. -∞	Histogram
Age	17	38.5817	?	90	13.6413	0.5587	-0.1665	4	0	0	17 90
FinalWeight	12,285	189,778.3665	?	1,484,705	105,549.9777	1.447	6.2188	0	0	0	12.285 1.484.705
EducationNum	1	10.0807	?	16	2.5728	-0.3117	0.6233	2	0	0	1 16
CapitalGain	0.0	1,077.6488	?	99,999	7,385.2921	11.9538	154.7994	0	0	0	0 99.999
CapitalLoss	0.0	87.3038	?	4,356	402.9602	4.5946	20.3768	0	0	0	0 4.356
HoursPerWeek	1	40.4375	?	99	12.348	0.2276	2.9161	3	0	0	1 99

Figure 1: Descriptive statistics and distributions of numerical variables



Workclass	Occupation
No. missings: 1836	**No. missings:** 1843
Top 20:	**Top 20:**
Private : 22696	Prof-specialty : 4140
Self-emp-not-inc : 2541	Craft-repair : 4099
Local-gov : 2093	Exec-managerial : 4066
? : 1836	Adm-clerical : 3770
State-gov : 1298	Sales : 3650
Self-emp-inc : 1116	Other-service : 3295
Federal-gov : 960	Machine-op-inspct : 2002
Without-pay : 14	? : 1843
Never-worked : 7	Transport-moving : 1597
	Handlers-cleaners : 1370
	Farming-fishing : 994
	Tech-support : 928
	Protective-serv : 649
	Priv-house-serv : 149
	Armed-Forces : 9

Figure 2: Frequency analysis of nominal variables

Based on the initial exploratory data analysis (EDA) using the Statistics views, several key insights and data quality issues were identified:

- Identification of Standard and Hidden Missing Values: The statistical summary revealed two types of missing data. First, standard missing values were detected in numerical columns, still in small quantities (Age: 4, EducationNum: 2, HoursPerWeek: 3). Second, and more importantly, a detailed inspection of the categorical variables revealed a large number of "hidden" missing values. The features most heavily affected by these unknown values are Workclass (1,836 instances) and Occupation (1,843 instances). These rows must be addressed through imputation or filtering in the subsequent data preparation phase.
- Target Variable Imbalance: This dataset indicates that the target variable (IncomeLevel) is highly imbalanced, with the majority of individuals falling into the "<=50K" category.

These results show the necessary steps for the upcoming Data Preparation phase, ensuring that missing values are handled properly and the data is optimally formatted for algorithm training.

4. Data Preparation

The Data Preparation phase covers all activities required to construct the final dataset from the initial raw data. Based on the insights gained during the Data Understanding phase, a sequence of preprocessing steps was executed within KNIME to ensure data quality, remove noise, and prepare a balanced dataset for model training.

- Data Filtering and Cleaning: Initial data reduction was performed using the Row Filter and Column Filter nodes. These nodes were utilized to eliminate irrelevant attributes (FinalWeight and Education) and filter out specific noisy records that do not contribute to the predictive power of the model. Following this, a Rule Engine node was implemented to standardize data entries(“United-States” to “US” and for other nations “Non-US”).
- Missing Value Imputation: To resolve the missing data issue identified during the exploratory analysis, the Missing Value node was used.

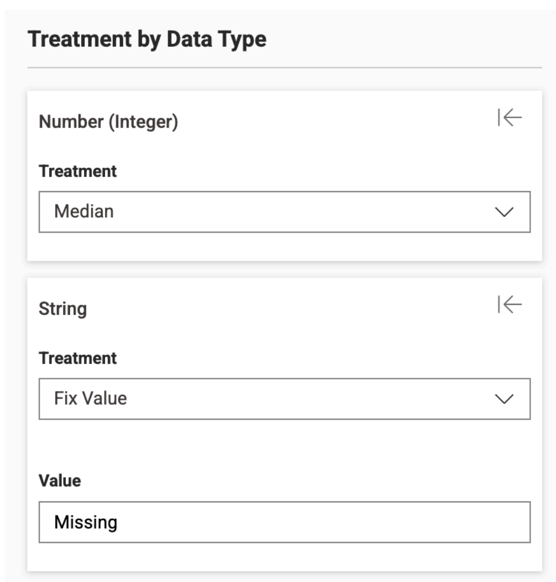


Figure 3: Configuration of the Missing Value node for data imputation

A specific strategy was applied based on the data type: missing numerical values (such as Age) were imputed using the median. For missing categorical values (such as Workclass and Occupation), a fixed value of "Missing" was assigned to replace the unknown entries, ensuring that all records remained consistent for the modeling stage.

- Visual Enhancement: A Color Manager node was applied directly to the target variable (IncomeLevel). This node assigns green for higher income and red to less income.
- Data Partitioning: The preprocessed dataset was split into training and testing subsets using the Table Partitioner node. A standard split (70% training and 30% testing) was applied.
- Data Balancing: As discovered during the initial data exploration, the target variable was highly imbalanced. To prevent the models from developing a bias toward the majority class ("<=50K"), an Equal Size Sampling node was applied exclusively to the training dataset. This under-sampling technique reduces the majority class to match the size of the minority class, providing a perfectly balanced distribution for model training while preserving the natural, real-world distribution of the test set for accurate evaluation.

5. Modeling

The following figure illustrates the complete KNIME analytical workflow developed for this project. It encompasses the entire end-to-end process, from data ingestion and cleaning to model training and performance evaluation.

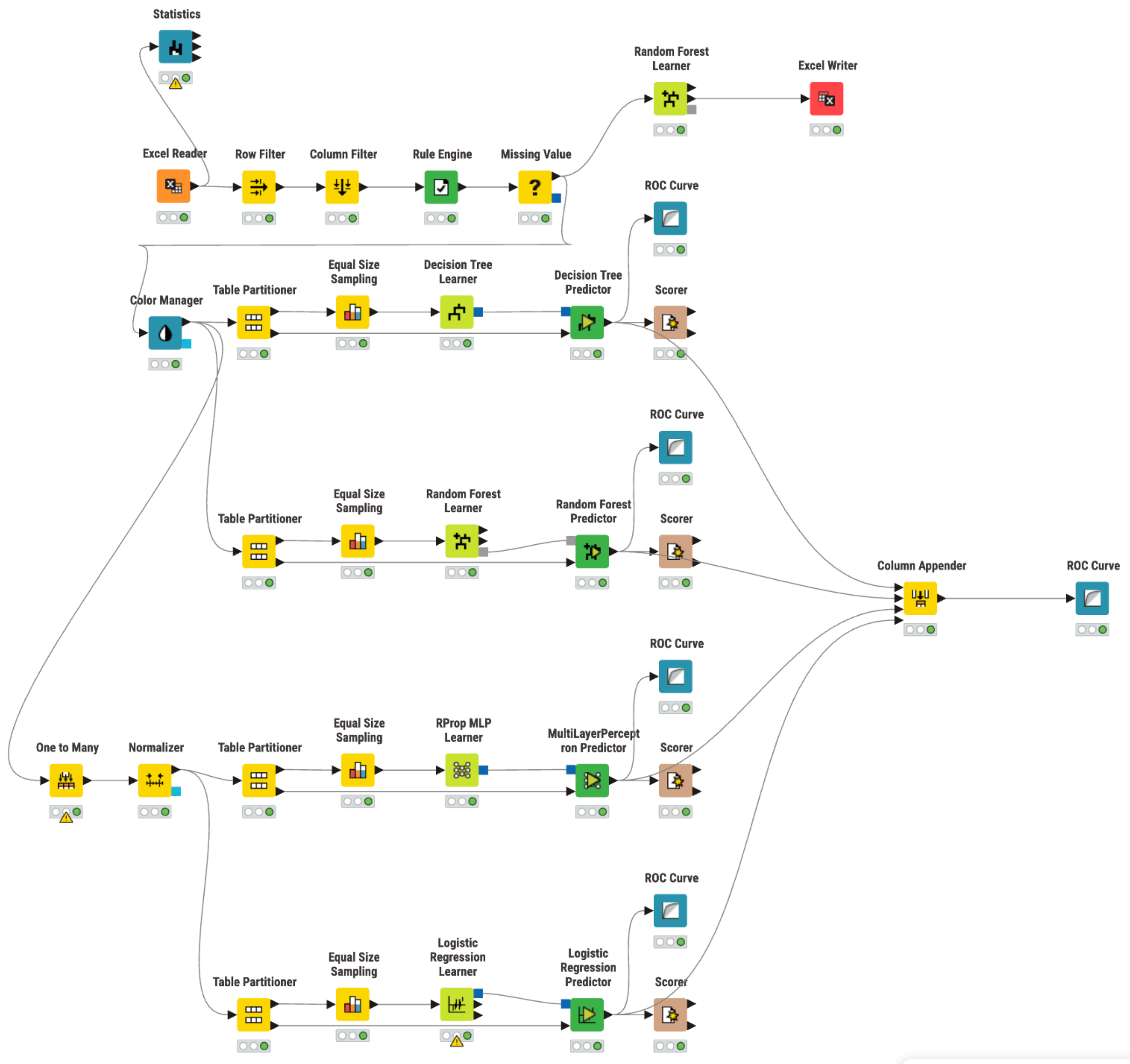


Figure 4: Overview of the complete final KNIME workflow for income classification

In this phase, four different machine learning models were built and trained: Decision Trees, Random Forest, Artificial Neural Networks - MLP and Logistic Regression. The training process followed two different paths based on the model requirements:

Tree Models: Decision Trees and Random Forest models were trained using the balanced dataset directly, as these algorithms can handle different types of data without extra scaling.

Mathematical Models: Logistic Regression and Artificial Neural Networks - MLP require numerical and scaled data. Therefore, the "One to Many" node was used to turn text data into numbers, and the "Normalizer" node was used to scale all values between 0 and 1.

All models were trained on the balanced data and then tested on the separate test set to see how well they predict real-world cases.

6. Evaluation

The models were evaluated using four main metrics: Accuracy, Sensitivity, Specificity, and ROC (AUC) values. The results were taken from the Scorer nodes of each model.

6.1. Performance Results

The following images show the detailed performance results for each algorithm:

#	RowID	TruePositiv... Number (Inte...	FalsePositi... Number (Inte...	TrueNegati... Number (Inte...	FalseNegati... Number (Inte...	Recall Number (Flo...	Precision Number (Flo...	Sensitivity Number (Flo...	Specificity Number (Flo...	F-measure Number (Flo...	Accuracy Number (Flo...	Cohen's kap... Number (Flo...
1	No	5649	431	1921	1766	0.762	0.929	0.762	0.817	0.837	?	?
2	Yes	1921	1766	5649	431	0.817	0.521	0.817	0.762	0.636	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.775	0.485

Figure 5: Scorer results for Decision Tree Classifier

#	RowID	TruePositiv... Number (Inte...	FalsePositi... Number (Inte...	TrueNegati... Number (Inte...	FalseNegati... Number (Inte...	Recall Number (Flo...	Precision Number (Flo...	Sensitivity Number (Flo...	Specificity Number (Flo...	F-measure Number (Flo...	Accuracy Number (Flo...	Cohen's kap... Number (Flo...
1	No	5990	334	2018	1425	0.808	0.947	0.808	0.858	0.872	?	?
2	Yes	2018	1425	5990	334	0.858	0.586	0.858	0.808	0.696	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.82	0.575

Figure 6: Scorer results for Random Forest Classifier

#	RowID	TruePositiv... Number (Inte...	FalsePositi... Number (Inte...	TrueNegati... Number (Inte...	FalseNegati... Number (Inte...	Recall Number (Flo...	Precision Number (Flo...	Sensitivity Number (Flo...	Specificity Number (Flo...	F-measure Number (Flo...	Accuracy Number (Flo...	Cohen's kap... Number (Flo...
1	No	5825	311	2041	1590	0.786	0.949	0.786	0.868	0.86	?	?
2	Yes	2041	1590	5825	311	0.868	0.562	0.868	0.786	0.682	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.805	0.551

Figure 7: Scorer results for Artificial Neural Networks – MLP

#	RowID	TruePo... Number (Inte...	FalsePositi... Number (Inte...	TrueNegati... Number (Inte...	FalseNegati... Number (Inte...	Recall Number (Flo...	Precision Number (Flo...	Sensitivity Number (Flo...	Specificity Number (Flo...	F-measure Number (Flo...	Accuracy Number (Flo...	Cohen's kap... Number (Flo...
1	No	5882	336	2016	1533	0.793	0.946	0.793	0.857	0.863	?	?
2	Yes	2016	1533	5882	336	0.857	0.568	0.857	0.793	0.683	?	?
3	Overall	?	?	?	?	?	?	?	?	?	0.809	0.554

Figure 8: Scorer results for Logistic Regression

The table below summarizes these results for a clear comparison:

Model	Accuracy	Sensitivity	Specificity	ROC (AUC)
Decision Tree	0.775	0.817	0.762	0.828
Logistic Regression	0.809	0.857	0.793	0.907
Neural Network (MLP)	0.805	0.868	0.786	0.909
Random Forest	0.820	0.858	0.808	0.909

Table 1: Summary of model performance metrics

According to the results, Random Forest is the most successful model with 82% accuracy and 0.909 AUC. While the Neural Network (MLP) also shows high performance, Random Forest provides a more balanced prediction across all categories.

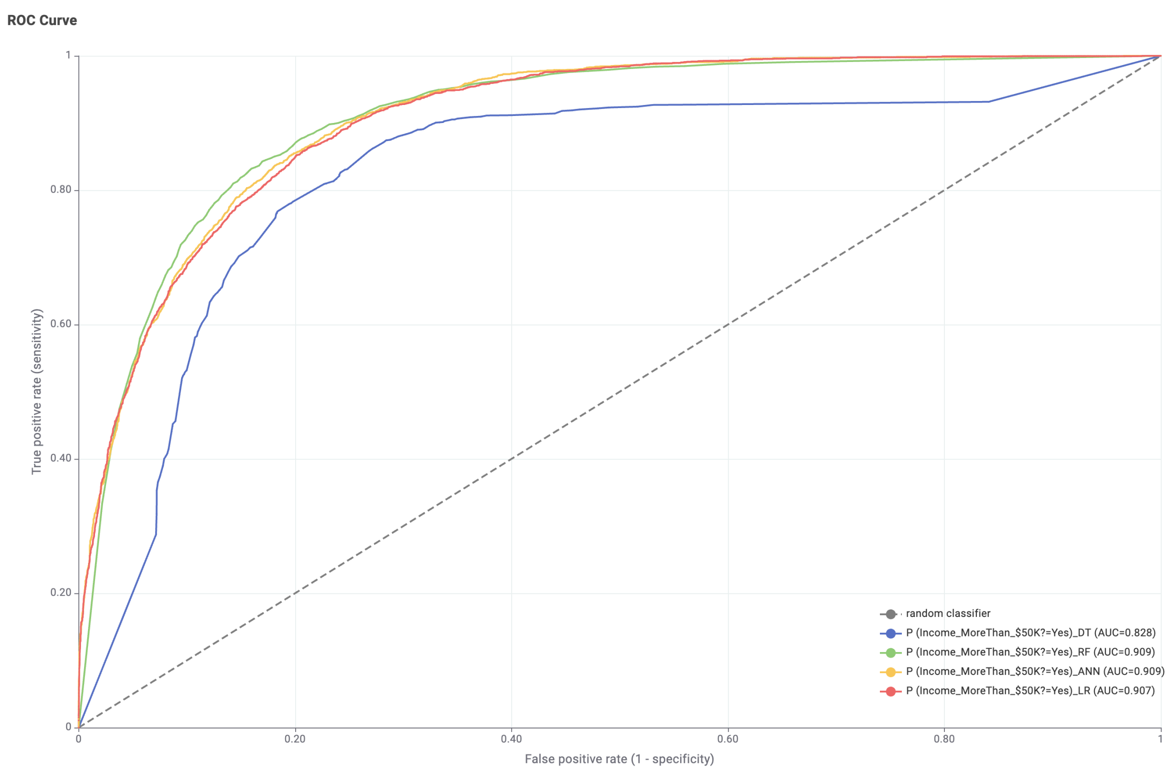


Figure 9: Combined ROC Chart comparing the predictive capabilities of all models

6.2. Variable Importance and Decision Tree Splits

Understanding which factors affect income

Figure 10: First three levels of the Decision Tree model

The Decision Tree shows that "Relationship" and "Capital Gain" are the first variables used to split the data.

RowID	Wsum
NativeCountry	0,001086
Race	0,00512
Gender	0,022047
HoursPerWeek	0,100721
Workclass	0,137881
Occupation	0,219752
EducationNum	0,299146
Age	0,429117
CapitalLoss	0,574986
MaritalStatus	0,770259
Relationship	0,826758
CapitalGain	1,083273

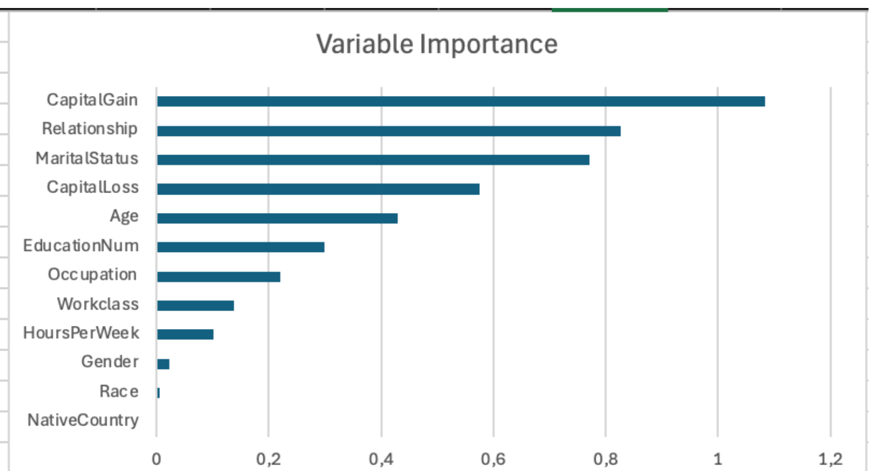


Figure 11: Variable Importance results from Random Forest.

The Variable Importance graph confirms these findings. Capital Gain is clearly the most important variable in predicting if someone earns more than $50,000. Other important factors include Relationship, Marital Status, and Age.

7. Summary and Conclusions

7.1. Model Recommendation for Business Use

Based on the extensive evaluation, the Random Forest model is recommended for deployment. If a financial institution, marketing agency, or government department were to use this system, they could rely on the Random Forest algorithm to accurately predict high-income individuals with an 82% accuracy rate and a 0.909 ROC value. The model is highly robust, balanced, and ready to classify new, unseen data.

7.2. Actionable Business Insights

The deployment of this predictive model provides valuable knowledge for decision-makers:

- Investment and Income: The most critical factor in predicting an income over $50,000 is Capital Gain. Therefore, people who have investment income are more likely to be high-income customers.
- Family Dynamics: Personal life structure, specifically Relationship and Marital Status, are the next most important factors.

7.3. Project Conclusion

This project completed all six phases of the CRISP-DM methodology using the KNIME Analytics Platform. The data was carefully explored, cleaned, and balanced. Missing values were handled. By comparing four different machine learning algorithms (Decision Trees, Random Forest, MLP, and Logistic Regression), the project successfully developed a reliable and highly accurate classification tool to predict income levels based on census data.